

Urban Flow: SaaS

Requirements

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1 Introduction

1.1 System Overview

Urban Flow is a cloud-based Software-as-a-Service (SaaS) platform designed to help city planners evaluate potential changes within an urban district before they are implemented in reality. It provides a simplified digital representation of a district, including its zones, facilities, and operational capacities, allowing planners to explore different planning scenarios in a controlled environment.

The system centralizes district information and enables planners to create and compare multiple “what-if” scenarios. Users can introduce proposed changes, such as temporary closures, capacity adjustments, or service modifications, and receive summarized impact indicators that highlight potential constraints or risks. By organizing planning data in a structured way, the system supports more informed and coordinated decision-making across departments.

At a high level, the system consists of:

- A user interface where planners define districts and create planning scenarios
- A set of backend components that manage district information, scenario definitions, and impact calculations
- A centralized data repository that stores all planning data and evaluation results

Together, these components provide a structured and centralized approach to district-level planning and scenario evaluation.

2 Functional Requirements

Describe the services and operations provided by the product, and the relationship of the product to its environment including users. In this portion the system functionality must be described completely and precisely to some extent. All requirements must be numbered using the following scheme:

- FR 1. The system shall provide a client interface through which authorized users interact with system functionality.
- FR 2. The system shall authenticate users using a valid username and password provided during login.
- FR 3. The system shall enforce role-based access control
 - FR 3.1. An Administrator shall have system management privileges.
 - FR 3.1.1. The Administrator shall be able to create, modify and delete user accounts.
 - FR 3.1.2. The Administrator shall be able to assign roles to users.
 - FR 3.1.3. The Administrator shall be able to create, modify, and delete districts, facilities, and services.
 - FR 3.2. A Planner shall have permissions to manage planning scenarios and district data.
 - FR 3.2.1. The Planner shall be able to view, create and modify districts.
 - FR 3.2.2. The Planner shall be able to view, create and modify services within districts.
 - FR 3.2.3. The Planner shall be able to add, modify and remove assets within a district.
 - FR 3.2.4. The Planner shall be able to modify asset attributes such as capacity, availability, or operational status.
 - FR 3.2.5. The Planner shall be able to view, create and modify planning scenarios.
- FR 4. The system shall allow Administrators and Planners to create a district by providing the district name, geographic boundary or location, and district description.
- FR 5. The system shall allow Administrators and Planners to define zones within a defined district.
- FR 6. The system shall allow Administrators and Planners to create, update, and delete district assets. An asset represents a physical community facility within a district, such as a park, gym, hospital, school, or recreational facility.
 - FR 6.1. The system shall allow Administrators and Planners to create an asset by providing asset name, type, location, capacity and operational status.
 - FR 6.2. Each asset shall be associated with a specific zone within a district.
- FR 7. The system shall allow a Planner to create a planning scenario.
 - FR 7.1. When creating a planning scenario, the system shall require the Planner to provide a scenario name, the target district, and a description of the proposed change.
 - FR 7.2. The system shall allow the Planner to define the entities affected by the scenario, including district assets as well as infrastructure elements.
 - FR 7.3. The system shall allow the Planner to modify the operational state or attributes of affected entities within the scenario.
- FR 8. The system shall maintain version information for each scenario
- FR 9. The system shall allow scenarios to transition between draft and approved states.
- FR 10. The system shall allow a Planner to define operational changes within a scenario.
 - FR 10.1. When defining an operational change, the system shall require the Planner to specify the affected entity, the type of change, and the modified attribute value.
 - FR 10.2. A change shall include a defined effective date and time period.
- FR 11. The system shall prevent modification of changes once a scenario is approved.
- FR 12. The system shall allow a Planner to initiate the evaluation of a planning scenario.
- FR 13. The system shall analyze the impact of the scenario on affected district assets and roads.
- FR 14. The system shall allow the Planner to view the evaluation results for the scenario.
- FR 15. The system shall compute rule-based indicators estimating the impact of a scenario on district asset capacity utilization and road traffic conditions.

Non-Functional Requirements

- NFR 1. The system shall be available at least 99.99% within the whole year.
- NFR 2. The database shall be accessible only by the system services.
- NFR 3. The system services shall provide access to the provided interfaces only through http or https protocols.
- NFR 4. Specify the private and the public end points within the application logic that is suitable by the proposed design.
- NFR 5. The system must handle over 10,000 requests at a time by scaling out the needed resources based on demand.
- NFR 6. The system shall have a recovery point objective of less than 1 day.
- NFR 7. The system shall have a recovery time objective of less than 2 seconds.